

Milesight UG6x API Documentation

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1.API Authentication

1.1 Login

Using UG6x username and password for authentication.

Method: POST

Request Address

<https://{gatewayIP}:8080/api/internal/login>

Request Parameters

Name	Type	Description	Default
username	string	The UG6x API's username.	apiuser
password	string	The UG6x API's password.	password

Response Parameters

Name	Type	Description
jwt	String	Token used for subsequent requests.
Request successful		
The response code is 200		
Request failed		
error	String	Error code.

Examples

Request Example

```
curl -X POST --header 'Content-Type: application/json' --header 'Accept: application/json' -d
'{"password": "password", "username": "apiuser"}' --insecure
https://192.168.23.164:8080/api/internal/login
```

Response Example

```
{"jwt":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsbnJhLWZwcC1zZXJ2ZXIiLCJleHAiOiJlE2M
Tg0MDIxNTQsImVzcyI6ImxvcmEtYXBwLXNlcnZlciIsIm5iZiI6MTYxODMxNTc1NCwic3VlJoidXNlc
iIsInVzZXJuYV11IjoieYXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE"
}
```

1.2 Change User Password

Method: PUT

Request Address

<https://{gatewayIP}:8080/api/users/{username}/password>

Request Parameters

Name	Importance	Type	Description
username	Required	string	The name of the user for which to update the password.

Response Parameters

Request successful			
The response code is 200.			

Request failed
Error code.

Examples**Request Example**

```
curl -X PUT --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsb3JhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlY2MTg0MDIxNTQsImZcyI6ImxvcmEtYXBwLXNlcnZlcil5iZil6MTYxODMxNTc1NCwic3Vljb2NlciIsInVzZXJuYW1lIjoieXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE' -d '{"password": "123456"}' --insecure https://192.168.23.164:8080/api/users/apiuser/password
```

Response Example

```
{}
```

2. Application

2.1 Get All Available Applications

Method: GET

Request Address

https://{gatewayIP}:8080/api/applications

Request Parameters

Name	Importance	Type	Description
limit	Required	string	Max number of applications to return in the result-test.
offset	Required	string	Offset in the result-set (for pagination).

Response Parameters

Name	Type	Description
description	String	The the description of the application profile already created.
ID	String	The ID of the application profile already created.
name	String	The name of the application profile already created.
payloadCodec	String	The payload codec of the application profile already created.
payloadDecoderScript	String	The function of payload decoder.
payloadEncoderScript	String	The function of payload encoder.
totalCount	String	The total number of the application profile already created.
Request successful		
The response code is 200.		
Request failed		
error	String	Error code.

Note: the API token is valid in 24 hours.

Examples**Request Example**

```
curl -X GET --header 'Accept: application/json' --header 'Authorization: Bearer
```

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsbn3JhLWFwcC1zZXJ2ZXIiLCJleHAiOiJlE2MTg0MDIxNTQsImZcyI6ImxvcnEtYXBwLXNlcnZlcilsm5iZil6MTYxODMxNTc1NCwic3ViljoidXNlciIsInVzZXJuYW1lIjoieYXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE'
--insecure https://192.168.23.164:8080/api/applications?limit=10&offset=0
```

Response Example

```
{
  "apps":[
    {
      "id":"1",
      "name":"test",
      "description":"test",
      "payloadCodec":"",
      "payloadEncoderScript":"// Encode encodes the given object into an array
of bytes.\n//   - fPort contains the LoRaWAN fPort number\n//   - obj is an
object, e.g. {\"temperature\": 22.5}\n// The function must return an array
of bytes, e.g. [225, 230, 255, 0]\nfunction Encode(fPort, obj) {\n  return
[];\n}","payloadDecoderScript":"// Decode decodes an array of bytes into an
object.\n//   - fPort contains the LoRaWAN fPort number\n//   - bytes is an
array of bytes, e.g. [225, 230, 255, 0]\n// The function must return an object,
e.g. {\"temperature\": 22.5}\nfunction Decode(fPort, bytes) {\n  return
{};\n}"
    },
    {
      "id":"2",
      "name":"test2",
      "description":"nancy",
      "payloadCodec":"",
      "payloadEncoderScript":"// Encode encodes the given object into an array of
bytes.\n//   - fPort contains the LoRaWAN fPort number\n//   - obj is an
object, e.g. {\"temperature\": 22.5}\n// The function must return an array of
bytes, e.g. [225, 230, 255, 0]\nfunction Encode(fPort, obj) {\n  return
[];\n}","payloadDecoderScript":"// Decode decodes an array of bytes into an
object.\n//   - fPort contains the LoRaWAN fPort number\n//   - bytes is an
array of bytes, e.g. [225, 230, 255, 0]\n// The function must return an object,
e.g. {\"temperature\": 22.5}\nfunction Decode(fPort, bytes) {\n  return
{};\n}"
    }
  ],
  "totalCount":"2"
}
```

2.2 Get Application by Name

Method: GET

Request Address

https://{gatewayIP}:8080/api/applications/{name}

Request Parameters

Name	Importance	Type	Description
name	Required	string	The name of the application profile already created.

Response Parameters Response Parameters

Name	Type	Description	
description	String	The the description of the application profile already created.	
id	String	The ID of the application profile already created.	
name	String	The name of the application profile already created	
payloadCodec	String	The payload codec of the application profile already created.	
payloadDecoderScript	String	The function of payload decoder.	
payloadEncoderScript	String	The function of payload encoder.	
Request successful			
The response code is 200.			
Request failed			
error	Optional	String	Error code

Note that the API token is valid in 24 hours.

Examples

Request Example

```
curl -X GET --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsbnJhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlY2MTg0MDIxNTQsImVzcyI6ImxvcmtYXBwLXNlcnZlciIsIm5iZiI6MTYxODMxNTc1NCwic3VlJoidXNlciIsInVzZXJuYV1lJoiYXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE' --insecure https://192.168.23.164:8080/api/applications/test
```

Response Example

```
{
  "id": "1",
  "name": "test",
  "description": "test",
  "payloadCodec": "",
  "payloadEncoderScript": "// Encode encodes the given object into an array of bytes.\n// - fPort contains the LoRaWAN fPort number\n// - obj is an object, e.g. {\"temperature\": 22.5}\n// The function must return an array of bytes, e.g. [225, 230, 255, 0]\nfunction Encode(fPort, obj) {\n  return [];\n}",
  "payloadDecoderScript": "// Decode decodes an array of bytes into an object.\n// - fPort contains the LoRaWAN fPort number\n// - bytes is an array of bytes, e.g. [225, 230, 255, 0]\n// The function must return an object, e.g. {\"temperature\": 22.5}\nfunction Decode(fPort, bytes) {\n  return {};\n}"
}
```

3. Device

3.1 Get All Available Devices

Method: GET

Request Address

https://{gatewayIP}:8080/api/devices

Request Parameters

Name	Importance	Type	Description
limit	Required	string	Max number of applications to return in the result-test.
offset	Required	string	Offset in the result-set (for pagination).

Response Parameters

Name	Type	Description
appKey	String	Application Key.
appName	String	The name of the selected application.
appSKey	String	Network Session Key.
applicationID	String	The ID of the selected application.
description	String	The description of this device.
devAddr	String	Device Address.
devEUI	String	Device EUI.
fCntDown	String	Downlink Frame-counter
fCntUp	String	Uplink Frame-counter
lastSeenAt	String	Show the time of last packet transmitted.
name	String	The name of this device.
nwkSKey	String	Network Session Key.
profileName	String	The name of the selected profile.
totalCount	String	The total number of the application profile already created.
Request successful.		
The response code is 200.		
Request failed		
error	String	Error code.

Note that the API token is valid in 24 hours.

Examples

Request Example

```
curl -X GET --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsbnJhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlY2MTg0MDIxNTQsImZcyLl6ImxvcmEtYXBwLXNlcnZlcisiIm5iZil6MTYxODMxNTc1NCwic3VlIjoiaXNlcilslVzZXJlYXNlcil9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE' --insecure https://192.168.23.164:8080/api/devices?limit=100&offset=0
```

Response Example

```
{
  "devices": [
```

```

{
  "devEUI":"24e1245657899999",
  "name":"EM500-udl",
  "applicationID":"1",
  "appName":"test",
  "description":"nancy",
  "profileName":"test",
  "fCntUp":0,
  "fCntDown":0,
  "appKey":"5572404c696e6b4c6f52613230313823",
  "devAddr":"",
  "appSKey":"",
  "nwksKey":"",
  "lastSeenAt":"-"
},
{
  "devEUI":"24e124136a473211",
  "name":"EM300-TH", "applicationID":"1",
  "appName":"test",
  "description":"nancy",
  "profileName":"test",
  "fCntUp":10,
  "fCntDown":4,
  "appKey":"5572404c696e6b4c6f52613230313823",
  "devAddr":"062a4663",
  "appSKey":"3498d0023ed43b589a687f80ce4bbbe0",
  "nwksKey":"2664d201f258808f7c9b97ac24b18b87",
  "lastSeenAt":"3 hours ago"
}
],
"totalCount":"2"
}

```

3.2 Get Device by Name

Method: GET

Request Address

<https://{gatewayIP}:8080/api/devices/{devName}>

Request Parameters

Name	Importance	Type	Description
devName	Required	string	The name of the device.

Response Parameters

Name	Type	Description
appKey	String	Application Key.

appName	String	The name of the selected application.
appSKey	String	Network Session Key.
applicationID	String	The ID of the selected application.
description	String	The description of this device.
devAddr	String	Device Address.
devEUI	String	Device EUI.
fCntDown	String	Downlink Frame-counter
fCntUp	String	Uplink Frame-counter
lastSeenAt	String	Show the time of last packet transmited.
name	String	The name of this device.
nwkSKey	String	Network Session Key.
profileName	String	The name of the selected profile.
Request successful		
The response code is 200.		
Request failed		
error	String	Error code.

Note that the API token is valid in 24 hours.

Examples

Request Example

```
curl -X GET --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsbnJhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlE2MTg0MDIxNTQsImZlcyI6ImxvcmEtYXBwLXNlcnZlcilsm5iZil6MTYxODMxNTc1NCwic3ViIjoidXNlciIsInVzZXJuYVw1IjoiYXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE' --insecure https://192.168.23.164:8080/api/devices/EM300-TH
```

Response Example

```
{
  "devEUI":"24e124136a473211",
  "name":"EM300-TH",
  "applicationID":"1",
  "appName":"test",
  "description":"nancy",
  "profileName":"test",
  "fCntUp":10,
  "fCntDown":4,
  "appKey":"5572404c696e6b4c6f52613230313823",
  "devAddr":"062a4663",
  "appSKey":"3498d0023ed43b589a687f80ce4bbbe0",
  "nwkSKey":"2664d201f258808f7c9b97ac24b18b87",
  "lastSeenAt":"3 hours ago"
}
```

3.3 Device Uplink Data

3.3.1 Get all devices uplink items in the device-queue

It's will return a HTTP Streaming to get device uplink data.

Method: GET

Request Address

`https://{gatewayIP}:8080/api/urpackets`

Response Parameters

Name	Type	Description
payloadJSON	String	
Request successful		
The response code is 200.		
Request failed		
error	String	Error code.

Note that the API token is valid in 24 hours.

Examples

Request Example

```
curl -X GET --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsbnJhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlE2MTg0MDIxNTQsImZcyI6ImxvcmEtYXBwLXNlcnZlciIsIm5iZil6MTYxODMxNTc1NCwic3ViIjoidXNlciIsInVzZXJuYW1lIjoieXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE' --insecure https://192.168.23.164:8080/api/urpackets
```

Response Example

```
{
  "result":{
    "frequency": 904700000,
    "power": "-",
    "immediately": "-",
    "dataRate": "SF7BW125",
    "modulation": "LORA",
    "bandwidth": 125,
    "spreadFactor": 7,
    "bitRate": 0,
    "codeRate": "4/5",
    "gatewayMac": "24E124FFFEF1B88E",
    "timeSinceGPSEPOCH": "365358h11m20.702s",
    "timestamp": 3440432671,
    "rssi": "-59",
    "loraSNR": "14.0",
    "devEUI": "24E124412B221912",
    "time": "2021-09-10T14:11:02+08:00",
    "type": "UpCnf",
    "fCnt": 8166,
    "devAddr": "06555C1C",
```

```

    "adr": "true",
    "adrAckReq": "false",
    "ack": "false",
    "mic": "3e7bf86e",
    "appEUI": "24E124C0002A0001",
    "fPort": "85",
    "size": "6",
    "payloadBase64": "AwAABAAA",
    "payloadHex": "0300000040000",
    "enqueue": false,
    "classType": "Class A"
}

```

3.3.2 Get specific devices uplink items

Method: GET

Request Address

https://{gatewayIP}:8080/api/devices/{devEUI}/data

Request Parameters

Name	Importance	Type	Description
devEUI	Required	string	Hex encoded DevEUI string.

Response Parameters

Name	Type	Description
payloadJSON	String	
Request successful		
The response code is 200.		
Request failed		
error	String	Error code.

Note that the API token is valid in 24 hours.

Examples

Request Example

```
curl -X GET --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsb3JhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlY2MTg0MDIxNTQsImIzcyI6ImxvcmEtYXBwLXNlcnZlciIsIm5iZiI6MTYxODMxNTc1NCwic3ViIjoiaXNlciIsInVzZXJuYXW1IjoiYXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE' --insecure https://192.168.23.164:8080/api/devices/24e124136a473211/data
```

Response Example

```
{
  "result":{
    "type":"uplink",
    "payloadJSON":{"applicationID":"","applicationName":"","data":"","A2cXAQRoAAUAAA==","devEUI":"","devName":"","EM300-TH":"","fCnt":4,"fPort":85,"rxInfo":{"altitude":0,"latitude":0,"loRaSNR":14.8,"longitude":0,"mac":"","24e124ffef1
```

```

61f1\", \"name\": \"Local
Gateway\", \"rssi\": -34, \"time\": \"2021-04-13T13:16:23.36554Z\"}, \"tim
e\": \"2021-04-13T13:16:23.36554Z\", \"txInfo\": {\"adr\": false, \"codeRate
\": \"4/5\", \"dataRate\": {\"bandwidth\": 125, \"modulation\": \"LORA\", \"s
preadFactor\": 10}, \"frequency\": 472700000}}\"
}
}

```

3.4 Device Downlink Queue

3.4.1 Get all downlink items in the device-queue

Method: GET

Request Address

<https://{gatewayIP}:8080/api/devices/{devEUI}/queue>

Request Parameters

Name	Importance	Type	Description
devEUI	Required	string	The EUI of the device.

Response Parameters

Name	Type	Description
confirmed	boolean	Whether the payload must be sent as confirmed data down or not
data	String	Base 64 encoded data (plaintext, will be encrypted by LoRa Server)
devEUI	String	Hex encoded device eui string
fPort	String	FPort to use (must be > 0)
jsonObject	String	Decoded object (when application coded has been configured), when providing the 'jsonObject', you can omit 'data'
reference	String	Random reference (used on ack notification) (optional)
Request successful		
The response code is 200.		
Request failed		
error	Optional	String Error code.

Note that the API token is valid in 24 hours.

Examples

Request Example

```

curl -X GET --header 'Accept: application/json' --header 'Authorization: Bearer
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsb3JhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlMTYx
g0MDIxNTQsImVzcyI6ImxvcmlzYXBwLXNlcnZlciIsIm5iZiI6MTYxODMxNTc1NCwic3VlJoidXNlciI
sInVzZuYXV1IjoieXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE'
--insecure https://192.168.23.164:8080/api/devices/24e124136a473211/queue

```

Response Example

```

{
  "items": [
    {

```

```

        "devEUI":"24e124136a473211",
        "reference": "",
        "confirmed": true,
        "fPort": 85,
        "data": "/wMEEA==",
        "fCnt": 1
    },
    {
        "devEUI":"24e124136a473211",
        "reference": "",
        "confirmed": true,
        "fPort": 85,
        "data": "/wMEEA==",
        "fCnt": 2
    },
    {
        "devEUI":"24e124136a473211", "reference": "",
        "confirmed": true,
        "fPort": 85,
        "data": "/wMEEA==",
        "fCnt": 3
    },
    {
        "devEUI":"24e124136a473211",
        "reference": "",
        "confirmed": true,
        "fPort": 85,
        "data": "/wMEEA==",
        "fCnt": 4
    }
]
}

```

3.4.2 Add the given item to the device-queue

Method: POST

Request Address

<https://{gatewayIP}:8080/api/devices/{devEUI}/queue>

Request Parameters

Name	Importance	Type	Description
confirmed	Required	boolean	Whether the payload must be sent as confirmed data down or not
data	Required	String	Base 64 encoded data (plaintext, will be encrypted by LoRa Server)
devEUI	Required	String	Hex encoded device eui string

fPort	Required	String	FPort to use (must be > 0)
jsonObject	Required	String	Decoded object (when application coded has been configured),when providing the 'jsonObject', you can omit 'data'
reference	Required	String	Random reference (used on ack notification)(optional)

Response Parameters

Request successful			
The response code is 200.			
Request failed			
error	Optional	String	Error code

Note:

- The API token is valid in 24 hours.
- if device not activated, you will get such response:

```
{
  "error": "enqueue downlink payload error: get next downlink fcnt for device",
  "error": "rpc error: code = NotFound desc = object does not exist",
  "code": 13
}
```

Examples

Request Example

```
curl -X POST --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsb3JhLWFWcC1zZXJ2ZXIiLCJleHAiOiJlMTg0MDIxNTQ5ImlzcyI6ImxvcmEtYXBwLXNlcnZlcilIm5iZiI6MTYxODMxNTc1NCwic3ViljoidXNlciInVzZXJ1YW11IjoieXBpdXNlci9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6lEVRE' -d '{"confirmed":false,"data":{"UwEAAAEAAGQ=","devEU1":{"24e124136a473211"},"fPort":85}}' --insecure https://192.168.23.164:8080/api/devices/24e124136a473211/queue
```

Response Example

 $\{$

3.4.3 Flush the downlink device-queue

Method: DELETE

Request Address

<https://{gatewayIP}:8080/api/devices/{devEUI}/queue>

Request Parameters

Name	Importance	Type	Description
devEUI	Required	String	Hex encoded DevEUI string

Response Parameters

Request successful			
The response code is 200.			
Request failed			
error	Optional	String	Error code

Note that the API token is valid in 24 hours.

Examples

Request Example

```
curl -X DELETE --header 'Accept: application/json' --header 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiJsbnJhLWFWcC1zZXJ2ZXliLCJleHAiOiJlE2MTg0MDIxNTQsImZcyL6ImxvcmEtYXBwLXNlcnZlciIsIm5iZil6MTYxODMxNTc1NCwic3ViljoidXNlcilslVzZXJ uYW1lIjoieYXBpdXNlciJ9.vz_DMUMe559MRX6TH8oMR3qeUybHJ768g2CLU6IEVRE' --insecure https://192.168.23.164:8080/api/devices/24e124136a473211/queue
```

Response Example

```
{}
```

-End-